

STARKE COUNTY AP, IN, USA

WMO: 720593

Lat: 41.333N

Lon: 86.667W

Elev: 208

StdP: 98.85

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00187

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-15.2	-27.9	0.3	-12.9	-24.1	0.4	-10.9	11.7	0.9	10.8	-0.5	3.0	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	32.2	22.9	30.9	22.1	29.1	21.0	25.0	29.6	23.9	28.4	22.9	27.2	4.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.7	27.6	22.5	17.6	26.4	21.8	16.8	25.5	77.0	29.9	72.7	28.7	68.6	27.1	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.9	9.3	8.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-24.0	26.6	5.3	0.7	-27.8	27.1	-30.9	27.5	-33.9	27.9	-37.7	28.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-4.5	-2.2	4.0	9.3	16.0	20.8	21.6	20.9	17.8	11.3	4.2	-1.1
	DBStd	10.51	6.48	6.03	6.20	5.11	5.34	3.54	3.34	2.97	4.38	5.06	5.09	5.59
	HDD10.0	1661	451	343	205	74	11	0	0	0	1	45	185	347
	HDD18.3	3467	709	575	446	274	111	16	9	11	63	227	423	603
	CDD10.0	1622	0	2	19	53	195	323	361	337	234	84	12	2
	CDD18.3	385	0	0	2	3	37	89	111	89	47	8	0	0
	CDH23.3	3578	0	0	11	41	386	781	1071	794	421	75	0	0
	CDH26.7	1072	0	0	1	4	114	252	344	213	131	14	0	0
(14) Wind	WSAvg	3.6	4.3	4.2	4.3	4.6	3.9	3.1	2.2	2.0	2.4	3.5	4.0	4.2
(15) Precipitation	PrecAvg	1046	76	56	62	92	101	116	122	114	82	80	82	66
	PrecMax	1279	152	123	125	171	167	202	227	243	200	223	183	119
	PrecMin	836	33	28	22	24	30	58	65	47	29	31	22	23
	PrecStd	129	37	27	28	39	50	51	44	56	51	46	39	24
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	17.2	26.0	27.3	32.1	33.7	34.2	32.4	32.8	28.7	21.2	16.0
		MCWB	11.7	8.0	17.0	16.3	21.5	22.3	24.8	23.0	22.9	19.3	14.0	12.5
	2%	DB	10.2	12.7	20.2	24.1	29.8	31.3	32.1	31.0	30.8	25.4	17.4	12.2
		MCWB	8.5	8.5	13.6	14.4	19.1	21.2	24.3	22.4	21.6	17.9	12.7	9.6
	5%	DB	7.2	9.8	17.2	21.3	27.6	29.8	30.1	28.9	27.6	22.6	14.9	8.7
		MCWB	5.4	6.9	12.4	13.3	17.8	20.2	22.7	22.0	19.6	15.7	10.8	7.2
	10%	DB	3.6	6.0	13.8	18.0	25.4	27.6	28.0	27.5	25.8	20.0	12.3	6.2
		MCWB	1.4	3.0	9.4	11.6	17.5	19.6	21.4	20.6	18.4	15.0	9.0	4.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.4	11.3	17.7	18.2	22.9	25.5	26.4	25.5	24.4	21.7	15.7	13.8
		MCDB	12.8	13.4	24.0	23.7	29.4	30.8	31.4	30.0	30.1	26.2	18.4	15.1
	2%	WB	8.8	9.0	15.7	16.5	21.6	23.9	25.4	24.2	23.0	19.4	13.5	9.8
		MCDB	9.9	12.2	18.9	21.3	26.7	28.5	29.8	28.5	28.0	22.3	16.8	10.9
	5%	WB	5.1	6.4	12.8	14.9	20.3	22.5	24.3	23.1	21.9	17.4	11.3	6.8
		MCDB	7.3	10.3	15.9	19.5	24.8	26.7	28.6	27.0	25.9	19.8	14.3	8.2
	10%	WB	1.9	3.3	10.2	12.9	19.0	21.5	23.1	22.1	20.6	15.7	9.1	4.6
		MCDB	3.3	5.3	13.9	17.7	23.1	25.2	27.3	26.0	23.3	19.8	12.3	6.2
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.2	10.2	11.8	11.6	11.2	11.8	12.0	12.6	11.3	9.3	7.2
		MCDBR	10.6	12.7	13.8	15.7	14.1	13.1	12.7	13.3	13.7	14.4	12.6	10.3
	5% WB	MCWBR	9.2	9.9	9.0	8.7	6.4	5.6	5.8	6.2	6.2	7.4	8.4	9.1
		MCWBR	10.7	11.9	12.0	13.4	12.0	10.6	11.1	10.9	11.5	11.8	11.1	9.6
(40) Clear-Sky Solar Irradiance	taub		0.296	0.323	0.343	0.401	0.430	0.438	0.455	0.448	0.421	0.356	0.319	0.296
		taud	2.423	2.340	2.379	2.232	2.207	2.226	2.174	2.187	2.233	2.425	2.477	2.464
	Ebn at Noon		860	882	905	869	850	841	822	815	810	836	826	829
		Edh at Noon	79	100	108	135	141	139	145	139	124	91	75	70
(44) All-Sky Solar Radiation	RadAvg		1.58	2.42	3.58	4.65	5.29	6.05	6.08	5.32	4.41	2.90	1.83	1.32
	RadStd		0.15	0.30	0.29	0.40	0.48	0.38	0.32	0.31	0.36	0.29	0.24	0.13

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page